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# AI-Augmented Research

## Session 3.1: Authorship, Integrity, and the Limits of AI Assistance


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# Session 3.1 Overview

# AI Is Now Present in More Than Half of Peer Review

The integration of AI into scholarly communication has accelerated dramatically. This is not a future scenario; it is the current state of the scholarly record.

- *Nature* (2025): survey of 1,600 academics — **more than 50%** have used AI while peer reviewing manuscripts
- HAI Stanford (2024): **17.5%** of CS papers had AI-drafted content; **16.9%** of peer review text
- *Nature* (2026): tens of thousands of 2025 publications may contain invalid AI-generated references

The relevant question is not whether AI is present in scholarly communication, but how to engage with it responsibly.

# The Co-Author Mode: What AI Assistance in Writing Means

Zyphur (2026) identifies the **Co-author mode** as one of four appropriate AI-use modes. In this mode, AI assists with text that the researcher generated:

- Language polishing
- Register conversion (e.g., making academic prose more accessible)
- Length compression
- Structural reorganization

## The authorship boundary

The researcher must be able to **independently produce and defend every claim** in the manuscript. AI assistance on text the researcher cannot independently produce is not Co-author mode; it is substitution, and it crosses the authorship boundary.

# The Authorship Boundary: A Practical Definition

| Type of assistance | Example | Crosses boundary? |
|---|---|---|
| Language polishing | Improving sentence clarity | No |
| Register conversion | Making academic prose accessible | No |
| Length compression | Shortening a paragraph | No |
| Structural reorganization | Reordering sections | No |
| Argument generation | Producing claims the researcher cannot defend | Yes |
| Evidence fabrication | Generating citations the researcher has not read | Yes |
| Conclusion substitution | Producing interpretations the researcher cannot justify | Yes |

COPE (2023): authors are fully responsible for AI-assisted content; AI cannot be listed as author.
Habdija et al. (2024): ICMJE criteria — AI tools do not meet authorship requirements.

# Why Treating AI Use as a Plagiarism Question Is a Category Error

Many institutional policies frame AI use in writing as a plagiarism question: *did the researcher present AI-generated text as their own?*

Zyphur (2026) argues this framing is a **category error**. The relevant question is not ownership of text but **validity of the research**. A researcher who uses AI to generate claims they cannot defend has not merely plagiarized; they have published invalid research.

> The plagiarism frame misses the more serious problem. Validity, not ownership, is the primary concern.

Zyphur (2026): Part 1 — treating AI use as a plagiarism question misses the validity problem.

# Disclosure: What, When, and How

Disclosure of AI use in scholarly writing is now required by most major publishers. The general principle (COPE 2023, Blau et al. 2024):

1. Disclose the specific tool and version used
2. Describe the nature of the assistance (language editing, literature search, data extraction, etc.)
3. Confirm that all AI-generated content has been verified by the authors
4. Check the specific policy of the target journal before submission

Blau et al. (2024, *PNAS*): transparent disclosure and attribution as the first principle of scientific integrity.
Bozkurt (2024, *Open Praxis*): legal and ethical challenges of AI authorship and ownership.

# Publisher Policies: A Comparative Overview

| **Publisher/Journal** | **AI authorship** | **AI writing assistance** | **Disclosure** |
|---|---|---|---|
| *Nature* | Not permitted | Permitted with disclosure | Mandatory, in Methods |
| *Science* | Not permitted | Permitted with disclosure | Mandatory |
| JAMA/*Lancet* | Not permitted | Restricted; case-by-case | Mandatory |
| Elsevier | Not permitted | Permitted with disclosure | Mandatory, in cover letter |
| ACM | Not permitted | Permitted with disclosure | Mandatory |
| ICMJE guidelines | Does not meet criteria | Permitted with disclosure | Mandatory |

Habdija et al. (2024): ICMJE criteria and publisher policy landscape.

# Peer Review and Manuscript Evaluation: The Validator Mode

Zyphur (2026) identifies the **Validator mode** as the appropriate AI-use mode for peer review and manuscript evaluation.

In this mode, AI assists with critique:

- Identifying gaps in the argument
- Checking internal consistency
- Flagging methodological concerns

**Verification discipline for the Validator mode:** model-heterogeneity — run the critique through a model from a *different* AI family than the one that produced the original argument. Persistent critique across AI families is stronger evidence of a genuine problem.

Zyphur (2026): Competency 4 — model-heterogeneity in adversarial review.

# AI as a Brainstorm Prompt for Peer Review, Not a Completed Review

The appropriate role of AI in peer review is as a **brainstorm prompt**, not as a completed review.

AI can help a reviewer:

- Identify questions to ask
- Surface gaps to investigate
- Flag methodological concerns to raise

But the reviewer's own reading, domain expertise, and scholarly judgement are what constitute the review. A peer review that consists primarily of AI-generated text, without the reviewer's independent assessment, is **not a peer review**; it is an AI output with a human signature.

*Nature* (2025): the discipline for using AI responsibly in peer review is underdeveloped.

# Demonstration: AI-Assisted Manuscript Self-Review

## Using AI in Validator mode before submission

1. Upload the manuscript to Claude or ChatGPT
2. Ask: "What are the three weakest points in the argument of this paper?"
3. Evaluate each identified weakness: is it genuine or superficial?
4. Ask: "What methodological concerns would a rigorous reviewer raise about this paper?"
5. **Apply model-heterogeneity:** repeat steps 2–4 with a different AI system
6. Compare the two sets of critiques: where do they agree? Where do they disagree?
7. Address the genuine weaknesses before submission

**Verification step:** the researcher must independently evaluate whether each AI-identified weakness is substantive

# Verification, Hallucinations, and Reliability Assessment

The discovery-versus-verification distinction is most critical in the context of scholarly output.

- Walters & Wilder (2023, *Sci. Rep.*): **55%** of GPT-3.5 citations fabricated; **18%** of GPT-4 citations fabricated
- *Nature* (2026): hallucinated citations are polluting the scientific literature

> A manuscript that has been AI-polished but not human-verified is not ready for submission. Verification — checking every citation, factual claim, and inference — is the researcher's non-delegable responsibility.

# A Verification Protocol for AI-Assisted Manuscripts

## Five-step verification protocol

1. **Citation audit:** verify every citation against the primary source (DOI lookup, abstract check)
2. **Factual claim audit:** identify every factual claim and verify it against a primary source
3. **Inference audit:** confirm every inferential claim follows from the cited evidence
4. **Consistency check:** check that methods, results, and discussion are internally consistent
5. **Sycophancy check:** ask the AI to identify any claims in the manuscript that are overstated or unsupported

Document the results of each step and report them in the submission cover letter if required by the journal.

# AI for Visual Communication in Scholarly Output

AI tools for visual communication are now widely used in academic publishing. The scholarly obligations are:

1. **Accuracy:** the figure must accurately represent the data or the theoretical model
2. **Reproducibility:** the figure must be reproducible from the underlying data or the description
3. **Disclosure:** if AI generated the figure from scratch, disclose the tool and version
4. **Journal compliance:** check the target journal's policy on AI-assisted figures before submission

arXiv (2026): AI-generated figures in academic publishing — policies, tools, and disclosure requirements.

# Demonstration: AI-Assisted Figure Creation

## Creating a publication-ready conceptual diagram

1. Describe the theoretical model in plain language: "A mediation model where social media use affects academic performance through sleep quality"
2. Use Napkin.ai or Whimsical AI to generate a first-draft diagram
3. Evaluate: does the diagram accurately represent the mediation structure?
4. Iterate: provide corrections and regenerate
5. Export the final diagram and document: tool used, version, date, description provided
6. Check the target journal's policy: is disclosure required? What format?

**Verification step:** confirm that the diagram is consistent with the theoretical model described in the manuscript.

# Authorship, Attribution, and Ethical Concerns: A Structured Overview

AI use in research raises four distinct ethical concerns that must be addressed separately:

| Concern | Description | Current standard |
|---|---|---|
| Authorship | Can AI be listed as an author? | No — universally rejected by publishers |
| Attribution | Must AI assistance be disclosed? | Yes — mandatory at most major journals |
| Intellectual property | Who owns AI-generated content? | Contested; varies by jurisdiction |
| Research integrity | Does AI use affect validity? | Yes — validity is the researcher's responsibility |

Bozkurt (2024): legal and ethical challenges of AI authorship and ownership.
COPE (2023): authors are fully responsible for AI-assisted content.

# The Peer Review Confidentiality Problem

A specific ethical concern that has received insufficient attention: **uploading a manuscript under review to a third-party AI system may violate the confidentiality of the peer review process**.

The manuscript is shared with the reviewer in confidence; uploading it to a commercial AI system shares it with a third party. Many journals now explicitly prohibit this practice.

> Researchers acting as reviewers should check the journal's policy before using AI assistance in the review process.

# Session 3.1 Summary

## Three things to carry forward

1. The **authorship boundary** is defined by the researcher's ability to independently produce and defend every claim in the manuscript — not by the presence or absence of AI assistance
2. **Disclosure is mandatory** at most major journals and is warranted whenever AI materially shapes the research or the manuscript
3. **Verification** — checking every citation, factual claim, and inference — is the researcher's non-delegable responsibility, regardless of how much AI assistance was used

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# Questions and Discussion

Thank you!

Questions?

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